Photograph by BANDEEP SINGH



FRESH METAL Molten aluminium from recycled e-waste at a recycling plant in Roorkee

TREATING WASTE RIGHT

Given the alarming proportions the issue had assumed, the Union ministry of environment, forest and climate change (MoEFCC) passed stringent rules in 2016, setting targets for e-goods-producing companies to take the responsibility of routing old devices into recycling. Designated Extended Producer Responsibility (EPR), collection targets are devised as a ratio of total sales, and set through practices such as buyback. They were revised further in 2018. Between 2017-2018 and 2021-22, total annual collection targets set for 13 big consumer brands went up from 80,991 tonnes to 153,889 tonnes, according to a 2020 CSE report. These efforts led to a greater formalisation of the recycling process, as several Producer Responsibility Organisations (PROs, 77 at last count) set up collection centres on behalf of electronics manufacturers. They have now become a major link in routing e-waste from informal collectors to dedicated recycling units, whose number too has increased. The result? The industry estimates that 1.2 million tonnes of e-waste was recycled in India last year. Under EPR targets, 350,000 tonnes was collected and processed in FY21.

'It is a very exciting phase across verticals in waste management, given the government's carbon-neutral initiatives and the push climate change is bringing, in a big way," says A.L.N. Rao, who heads the e-waste division of the Material Recycling Association of India (MRAI), an industry body. He's also the CEO of e-waste recycling firm Exigo Recycling, and is on the NITI Aayog panel for a circular economy. "The country needs true recycling to happen and the urban mining concept to be successful for a circular economy to take off." New players have come into recycling and existing ones have invested in expansion, he points out, "because they know there is a big future in this industry." India's e-waste management is both a huge challenge and an opportunity. It starts with the urban consumer, who today must figure out what to do with their old device. "Are we willing to make that effort as consumers? Not really," says Siddharth Ghanshyam Singh, programme manager at CSE. Then there must be the e-waste from bulk consumers, including government departments. "There should be huge volumes," says P. Parthasarathy of E-Parisaraa in Bengaluru, which was the first e-waste recycling unit in India to get a central licence back in 2004. "Where is it going?"

Informal collectors tend to give households and bulk consumers a better price for their e-waste because of their low-cost methods. A 2018 report by Toxics Link, for instance, noted that all the informal recyclers in Delhi that they documented operated out of 'shabby and small rooms of residential or unauthorised colonies', none of them providing any occupational health safeguards to workers and with zero emission, effluent or waste processing. 'Waste was visibly dumped in the open or in community bins located close to drains, with the risk of contaminating soil, surface, sub-surface and ground water,' it had noted. "The only change that has happened over the years is that it has become a little more clandestine because they know regulations have come in," says Priti Mahesh. "Normally, people sell e-waste to scrap dealers who, in turn, dump the hazardous components in the city's river," says Karan Thakkar, who runs The Recycling Company, a 16,000 tonnes-per-annum recycling plant at Khopoli, around 70 kilometres from Mumbai. Thakkar, a first-generation entrepreneur, says it was initially difficult to convince people to hand over their e-waste, but the situation has improved gradually. "People call us voluntarily to collect their e-waste." Ironically enough, in a country where the rising junk level is clearly a problem, many formal recyclers complain that they don't get enough e-waste. Back in 2009 when he got his licence, Pasha of EwaRDD Recyclers re-